

STUDENT UTILITY PROGRAM

Objective / Aim

To design and implement a Student Utility Program in Java demonstrating variables and data types, input and output, conditional statements, loops, and parameterized methods.

Algorithm / Logic

1. Start the program.
2. Create a Scanner object to accept input from the user.
3. Read the student's name, roll number, and marks in three subjects.
4. Calculate total marks and percentage.
5. Display the student result and grade.
6. Display the Student Utility Menu.
7. Read the user's choice using a switch statement.
8. Perform the selected operation such as even/odd, largest number, grade, day, multiplication table, numbers from 1 to N, sum, Fibonacci series, factorial, prime checking, maximum of two numbers, or circle area.
9. Use methods for factorial, prime checking, maximum, circle area, and grade calculation.
10. Continue displaying the menu until the user selects Exit.
11. Close the Scanner.
12. Stop the program.

Steps for Execution

1. Create a Java project and create a class named StudentUtilityProgram.
2. Copy the given source code into the class.
3. Compile and run the program.
4. Enter the student name, roll number, and marks for three subjects.
5. View the total marks, percentage, and grade.
6. Select an option from the Student Utility Menu.
7. Enter the required input for the selected operation.
8. View the result displayed by the program.
9. Select option 13 to exit the program.

Source Code

```
import java.util.Scanner;

/**
 * Student Utility Program
 * Demonstrates:
 * 1. Variables and data types
 * 2. Input and output
 * 3. Conditional statements
 * 4. Loops
 * 5. Parameterized methods
 */public class StudentUtilityProgram {

    // ----- METHODS -----

}
```

```

    * Returns the factorial of a number.
    */
    public static long factorial(int n) {
        long result = 1;

        for (int i = 1; i <= n; i++) {
            result = result * i;
        }

        return result;
    }

    /**
     * Checks whether a number is prime.
     */
    public static boolean isPrime(int n) {
        if (n < 2) {
            return false;
        }

        for (int i = 2; i <= Math.sqrt(n); i++) {
            if (n % i == 0) {
                return false;
            }
        }

        return true;
    }

    /**
     * Returns the maximum of two numbers.
     */
    public static int maximum(int a, int b) {
        return (a > b) ? a : b;
    }

    /**
     * Calculates the area of a circle.
     */
    public static double circleArea(double radius) {
        return Math.PI * radius * radius;
    }

    /**
     * Displays the grade based on percentage.
     */
    public static void displayGrade(double percentage) {
        if (percentage >= 90) {
            System.out.println("Grade: A+");
        } else if (percentage >= 80) {
            System.out.println("Grade: A");
        } else if (percentage >= 70) {
            System.out.println("Grade: B");
        } else if (percentage >= 60) {
            System.out.println("Grade: C");
        } else if (percentage >= 50) {
            System.out.println("Grade: D");
        } else {
            System.out.println("Grade: F");
        }
    }

    // ----- MAIN METHOD -----

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Part A: Student Information
        System.out.println("===== STUDENT INFORMATION =====");

        System.out.print("Enter Student Name: ");
        String name = sc.nextLine();

        System.out.print("Enter Roll Number: ");
        int rollNumber = sc.nextInt();

        System.out.print("Enter marks in Subject 1: ");
        double mark1 = sc.nextDouble();

        System.out.print("Enter marks in Subject 2: ");
        double mark2 = sc.nextDouble();
        System.out.print("Enter marks in Subject 3: ");
    }

```

```

double mark3 = sc.nextDouble();

double total = mark1 + mark2 + mark3;
double percentage = total / 3;

System.out.println("\n===== STUDENT RESULT =====");
System.out.println("Name      : " + name);
System.out.println("Roll Number: " + rollNumber);
System.out.println("Total Marks: " + total);
System.out.println("Percentage : " + percentage + "%");

displayGrade(percentage);

// Menu-driven program
int choice;

do {
    System.out.println("\n===== STUDENT UTILITY MENU =====");
    System.out.println("1. Check Even or Odd");
    System.out.println("2. Find Largest of Three Numbers");
    System.out.println("3. Display Grade");
    System.out.println("4. Display Day of the Week");
    System.out.println("5. Multiplication Table");
    System.out.println("6. Display Numbers from 1 to N");
    System.out.println("7. Sum of First N Natural Numbers");
    System.out.println("8. Fibonacci Series");
    System.out.println("9. Factorial");
    System.out.println("10. Check Prime Number");
    System.out.println("11. Maximum of Two Numbers");
    System.out.println("12. Area of Circle");
    System.out.println("13. Exit");

    System.out.print("Enter your choice: ");
    choice = sc.nextInt();

    switch (choice) {

        case 1:
            // Even or Odd
            System.out.print("Enter a number: ");
            int number = sc.nextInt();

            if (number % 2 == 0) {
                System.out.println(number + " is Even.");
            } else {
                System.out.println(number + " is Odd.");
            }
            break;

        case 2:
            // Largest of three numbers
            System.out.print("Enter three numbers: ");
            int a = sc.nextInt();
            int b = sc.nextInt();
            int c = sc.nextInt();

            int largest;

            if (a >= b && a >= c) {
                largest = a;
            } else if (b >= a && b >= c) {
                largest = b;
            } else {
                largest = c;
            }

            System.out.println("Largest number = " + largest);
            break;

        case 3:
            // Grade based on percentage
            System.out.print("Enter percentage: ");
            double p = sc.nextDouble();

            displayGrade(p);
            break;

        case 4:
            // Day using switch
            System.out.print("Enter day number (1-7): ");
            int day = sc.nextInt();

            switch (day) {

```

```

        case 1:
            System.out.println("Monday");
            break;
        case 2:
            System.out.println("Tuesday");
            break;
        case 3:
            System.out.println("Wednesday");
            break;
        case 4:
            System.out.println("Thursday");
            break;
        case 5:
            System.out.println("Friday");
            break;
        case 6:
            System.out.println("Saturday");
            break;
        case 7:
            System.out.println("Sunday");
            break;
        default:
            System.out.println("Invalid day number.");
    }
    break;

case 5:
    // Multiplication table
    System.out.print("Enter a number: ");
    int tableNumber = sc.nextInt();

    for (int i = 1; i <= 10; i++) {
        System.out.println(
            tableNumber + " x " + i +
            " = " + (tableNumber * i)
        );
    }
    break;

case 6:
    // Numbers from 1 to N
    System.out.print("Enter N: ");
    int n = sc.nextInt();

    for (int i = 1; i <= n; i++) {
        System.out.print(i + " ");
    }
    System.out.println();
    break;

case 7:
    // Sum of first N natural numbers
    System.out.print("Enter N: ");
    int limit = sc.nextInt();

    int sum = 0;

    for (int i = 1; i <= limit; i++) {
        sum += i;
    }

    System.out.println(
        "Sum of first " + limit +
        " natural numbers = " + sum
    );
    break;

case 8:
    // Fibonacci series
    System.out.print("Enter number of terms: ");
    int terms = sc.nextInt();

    int first = 0;
    int second = 1;

    System.out.println("Fibonacci Series:");

    for (int i = 1; i <= terms; i++) {
        System.out.print(first + " ");

        int next = first + second;
        first = second;
        second = next;
    }

```

```

    }

    System.out.println();
    break;

case 9:
    // Factorial using method
    System.out.print("Enter a number: ");
    int factNumber = sc.nextInt();

    if (factNumber < 0) {
        System.out.println(
            "Factorial is not defined for negative numbers."
        );
    } else {
        System.out.println(
            "Factorial = " + factorial(factNumber)
        );
    }
    break;

case 10:
    // Prime number using method
    System.out.print("Enter a number: ");
    int primeNumber = sc.nextInt();

    if (isPrime(primeNumber)) {
        System.out.println(
            primeNumber + " is a Prime Number."
        );
    } else {
        System.out.println(
            primeNumber + " is not a Prime Number."
        );
    }
    break;

case 11:
    // Maximum of two numbers using method
    System.out.print("Enter two numbers: ");
    int x = sc.nextInt();
    int y = sc.nextInt();

    System.out.println(
        "Maximum = " + maximum(x, y)
    );
    break;

case 12:
    // Area of circle using method
    System.out.print("Enter radius: ");
    double radius = sc.nextDouble();

    System.out.println(
        "Area of Circle = " + circleArea(radius)
    );
    break;

case 13:
    System.out.println("Exiting program...");
    break;

default:
    System.out.println("Invalid choice.");
}

} while (choice != 13);

sc.close();
}
}

```

Explanation of the Program
Part A – Java Basics

The program accepts:

Student name using String

Roll number using int

Marks using double

The total and percentage are calculated using arithmetic operators.

```
double total = mark1 + mark2 + mark3;
double percentage = total / 3;
```

Input is obtained using the Scanner class.

```
Scanner sc = new Scanner(System.in);
String name = sc.nextLine();
int rollNumber = sc.nextInt();
```

Part B – Conditional Statements

The program demonstrates both if-else and switch.

For example, even/odd checking uses the modulus operator:

```
if (number % 2 == 0) {
    System.out.println("Even");
} else {
    System.out.println("Odd");
}
```

The switch statement is used to identify the day:

```
switch (day) {
    case 1:
        System.out.println("Monday");
        break;
    case 2:
        System.out.println("Tuesday");
        break;
    // ...
}
```

Part C – Loops

A for loop is used for the multiplication table:

```
for (int i = 1; i <= 10; i++) {
    System.out.println(tableNumber * i);
}
```

It is also used for the Fibonacci series:

```
int first = 0;
int second = 1;

for (int i = 1; i <= terms; i++) {
    System.out.print(first + " ");

    int next = first + second;
    first = second;
    second = next;
}
```

Part D – Parameterized Methods

Each required method accepts parameters and returns a result.

```
public static int maximum(int a, int b) {
    return (a > b) ? a : b;
}
```

Example method call:

```
int result = maximum(25,
```